**Republic of Tunisia**
**The Ministry of Higher Education and Scientific Research**

**The University of Carthage**

**The Higher Institute of Information and Communication Technologies**

END-OF-STUDY PROJECT REPORT

**Submitted in Partial Fulfillment of the Requirements for the**
BACHELOR DEGREE IN COMPUTER SCIENCE

**Field of Study : Software Engineering and Information Systems**

# Design and implementation of a network intrusion detection system

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Conducted within ITXTREE

Publicly defended on June 6, 2026 in front of the jury members:

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Academic Year: 2025-2026

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**Authorization of graduation project report submission:**

Professional Supervisor:

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# Dedications

I dedicate this End-of-Study project

**To my dear parents**

Who have supported and encouraged me every step of the way. This is for you, in honor of all the sacrifices you have made for me. I hope to have made you both proud and happy.

**To my dear sisters and brothers**

Who have been a great support and a source of joy in my life. I am grateful for your love and encouragement throughout this journey.

**To all my family and friends**

Thank you all for your support and encouragement throughout all this time. I am truly blessed to have you in my life.

**Amira BOUAFIF**

# Dedications

I dedicate this End-of-Study project

**To my dear parents**

**To my dear sisters**

**To all my family and friends**

**Hajer JEMAA**

# Acknowledgments

We would like to extend our sincere gratitude to all those who have supported us throughout the course of this End-of-Study project. Their guidance, encouragement, and support have been invaluable throughout this journey.

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we also extend our deep appreciation to the professors at the Higher Institute of Information and Communication Technologies for their invaluable support and teachings, which have greatly contributed to our academic journey.

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# General Introduction

# Chapter 1

# Project Context

## 1.1 Introduction

This chapter serves as an introduction to the project. We begin with a presentation of the company hosting the internship, a study of existing network intrusion detection systems. Then, we present the problem and the proposed solution. Finally, we end the chapter with a conclusion.

## 1.2 Presentation of the company

ITXTREE is an early-stage technology startup based in Sidi Rzig, Tunisia, focused on building a solid foundation for future digital solutions it is built on the core values of integrity, quality, curiosity and adaptability.

## 1.3 Study of the existing

## 1.4 Criticism of the existing

## 1.5 Proposed solution

Our goal is developing a network intrusion detection system that addresses the identified shortcomings through the following features:

## 1.6 Project Pipeline

## 1.7 Development Methodology

## 1.8 Conclusion

In this chapter we introduce the host company ITXTREE, study existing network intrusion detection systems, discuss their shortcomings and finally present a solution. In the next chapter we specify the system requirements.

# Chapter 2

# Requirements Specification

## 2.1 Introduction

## 2.2 Functional Requirements

## 2.3 Non-Functional Requirements

## 2.4 Actors Identification

Actors are external entities, users, organizations, or external systems that interact with the system to achieve one or more functional requirements, with each actor representing a specific role.

In the context of our project, we identify three primary actors illustrated in Figure 2.1

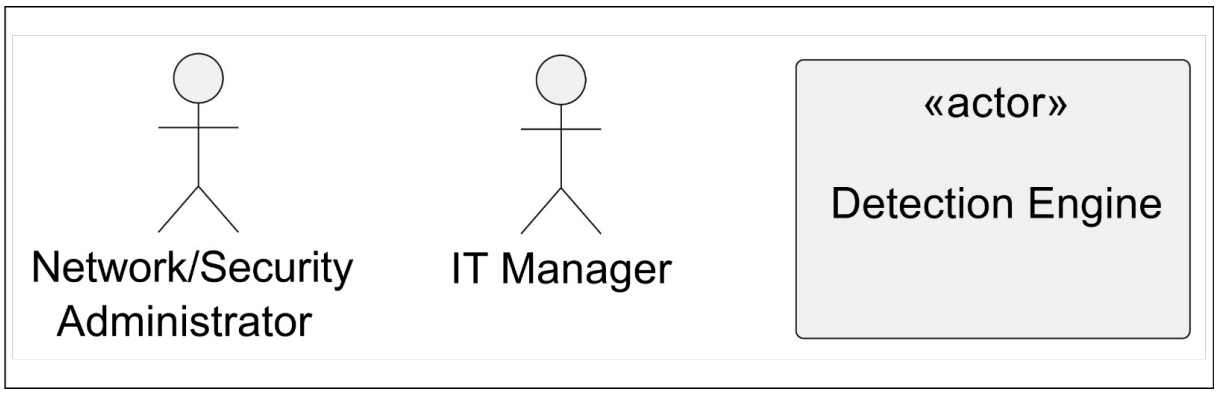


FIGURE 2.1 – Actors

Table 2.1 presents the roles of each actor in our system.

TABLE 2.1 – Roles of Actors

| Actor | Role |
|---|---|
| Network/Security Administrator | - Authenticate<br>- View Dashboard<br>- View Statistics Summary<br>- Trigger Network Analysis<br>- View Alerts List |
| It Manager | - Authenticate<br>- View Dashboard<br>- View Statistics Summary<br>- View Reports |
| Detection Engine | - Analyze Network Flow |

## 2.5 Use Case Diagram

A use case diagram is a visual representation of *how* users interact with the system and *what* a system does through different use cases. Figure 2.2 illustrates that diagram.

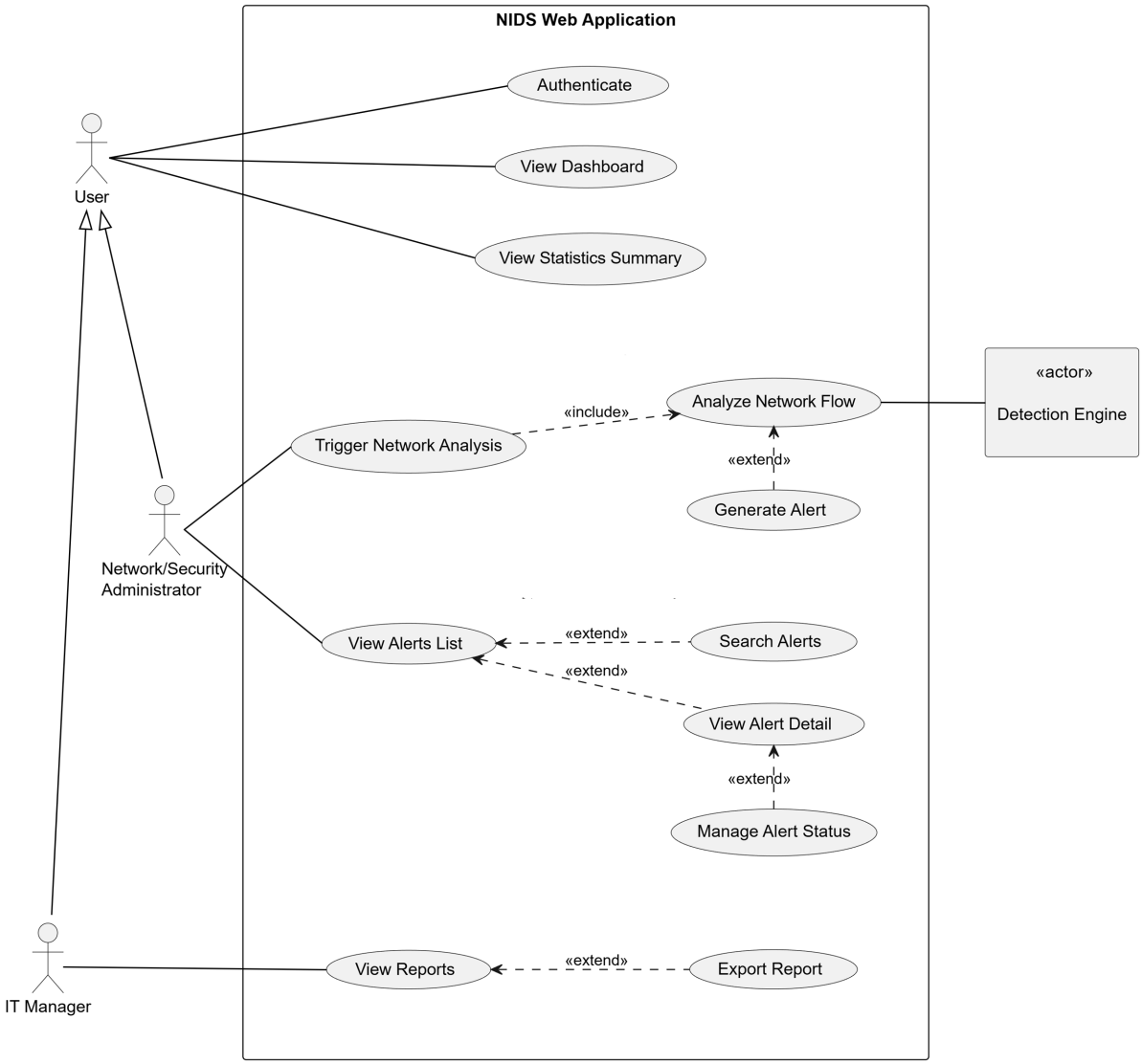


FIGURE 2.2 – Use Case Diagram

## 2.6 Conclusion

# Chapter 3

# Conceptual Design

## 3.1 Introduction

## 3.2 Sequence Diagrams

## 3.3 Class Diagram

## 3.4 Deployment Diagram

## 3.5 Conclusion

# Chapter 4

# Data Collection and Preparation

## 4.1 Introduction

## 4.2 Data Collection

## 4.3 Data Understanding

## 4.4 Data Preparation

## 4.5 Data Augmentation

## 4.6 Conclusion

# General Conclusion